

# Indexing All $(r, s)$ -Nucleus Decompositions

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## Abstract

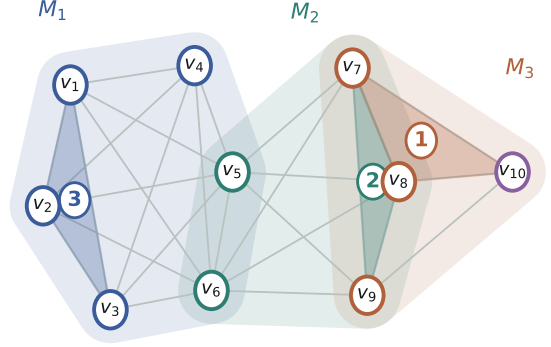
Dense subgraphs carry much of the meaning of a graph, and decomposing a graph into hierarchies of dense subgraphs is a fundamental task in graph mining. The  $(r, s)$ -nucleus decomposition is the clique-level member of this family, scoring every  $r$ -clique by its surrounding  $s$ -cliques. It subsumes both the  $k$ -core and the  $k$ -truss. The parameter pair  $(r, s)$  acts as a resolution, and analysts need many cells of the parameter plane rather than one. Existing algorithms compute one cell at a time and materialize every  $r$ -clique of that cell. A plane is therefore out of reach: a seven-million-edge web graph already contains more than  $10^{12}$  5-cliques. We observe that vertices belonging to exactly the same maximal cliques are interchangeable, and that the whole plane respects this symmetry. Based on this observation, we propose NSI, an index built on the clique quotient of the graph. Cliques collapse into patterns, support counts collapse into polynomial coefficients, and two transfer theorems reduce the whole plane to a single peeled cell. Wherever the closed form is certified, the index stores nothing and reconstructs the answer at query time. On real graphs the index is up to seven orders of magnitude smaller than the explicit table it replaces, and it answers point queries in tens of nanoseconds. Its builder outperforms the state-of-the-art single-cell algorithm by up to  $742\times$  in time and  $226\times$  in memory. A one-number test, computable before any index is built, tells in advance whether a graph is suitable.

## 1 Introduction

A graph represents entities as vertices and their pairwise relationships as edges. Groups of vertices that are densely connected to one another often carry the meaning of the whole network. For example, in collaboration networks, dense groups correspond to research teams working closely together [5]; in web graphs, they mark clusters of pages on one topic [4]; in biological networks, they capture protein complexes [15]; in financial and social platforms, they expose coordinated fraud rings [1, 8].

Decomposing a graph into a hierarchy of nested dense subgraphs is a fundamental task in graph mining. Representative applications include community detection [5, 16], densest-subgraph discovery [3], influence analysis [10, 12], and graph visualization [14]. In the literature, the task has been studied at three levels of granularity. The  $k$ -core peels vertices by degree [2, 20]; the  $k$ -truss peels edges by triangle count [21, 23]; the  $(r, s)$ -nucleus decomposition peels  $r$ -cliques by  $s$ -clique count [18]. The nucleus model subsumes the other two: the  $(1, 2)$ -nucleus is exactly the  $k$ -core, and the  $(2, 3)$ -nucleus is exactly the  $k$ -truss [18]. The parameter pair  $(r, s)$  therefore acts as a resolution, and every cell of the parameter plane reveals the graph’s dense structure at a different granularity [18, 19]. The state-of-the-art algorithm, CND [22], computes the decomposition of one cell at a time. In this paper, we aim to answer every cell of a bounded parameter plane from one index.

For integers  $r < s$ , consider subgraphs in which every  $r$ -clique is contained in at least  $k$   $s$ -cliques of the same subgraph. The nucleus value of an  $r$ -clique is the largest  $k$  for which such a subgraph



**Figure 1: An example graph  $G$  and three of its  $(3, 4)$ -nucleus values.**

contains it. Computing the cell  $(r, s)$  means computing this value for every  $r$ -clique of the graph.

*Example 1.1.* Consider the graph  $G$  in Figure 1, whose ten vertices form three maximal cliques:  $M_1$  with six vertices,  $M_2$  with five, and  $M_3$  with four. At the cell  $(3, 4)$ , the triangle  $\{v_1, v_2, v_3\}$  has nucleus value 3: it lies in the subgraph induced by  $M_1$ , in which every triangle is contained in at least three 4-cliques. The triangle  $\{v_7, v_8, v_9\}$  has value 2, and the triangle  $\{v_7, v_8, v_{10}\}$  has value 1. Other cells give other resolutions: at  $(3, 5)$  the value of  $\{v_1, v_2, v_3\}$  is still 3, while at  $(4, 5)$  the 4-clique  $\{v_1, v_2, v_3, v_4\}$  has value 2.

**Applications.** Many uses of the decomposition need several cells of the plane rather than one. *Multi-resolution exploration.* An analyst inspecting a dense region rarely knows the right resolution in advance. Low cells surface broad communities, while high cells isolate small, strongly coordinated groups [8, 18]. Exploration therefore issues a stream of queries across cells, and each query must return in interactive time. *Parameter selection.* Which cell best separates signal from noise differs across graphs and tasks [19]. Practitioners choose by probing: compute several cells, compare the resulting hierarchies, and keep the most informative one. Today every probe costs a full decomposition run. *Point queries at query time.* Downstream systems ask point questions: given one concrete group of vertices, how dense is its surrounding structure at each resolution [17]? Answering from the raw graph requires a fresh decomposition per question.

**Existing Methods and Key Limitation.** The state-of-the-art algorithm CND [22] enumerates every  $r$ -clique of the requested cell, counts its supporting  $s$ -cliques, and repeatedly removes the clique of minimum count. Careful counting structures let it obtain these counts without listing individual  $s$ -cliques, which made single-cell decomposition practical on graphs with millions of edges [22]. However, all existing algorithms share a fundamental limitation: they materialize *every  $r$ -clique of every cell they compute*. The number of  $r$ -cliques grows combinatorially with  $r$ : web-it-2004, a web graph

with 7.2 million edges, contains more than  $1.5 \times 10^{12}$  5-cliques. Even a single shallow cell is already prohibitive: on web-it-2004, CND needs 5,581 seconds and 101 GB of memory for the single cell (3, 4). Within a 1,800-second, 300 GB budget, CND finishes none of the nine web-it-2004 cells of our evaluation grid (Section 8).

This leads to the central research question of this paper: **Can one index answer the nucleus value of every  $r$ -clique, at every cell of a bounded plane, without ever materializing the plane?**

**Challenges.** Answering the plane from one structure faces three obstacles. Astronomically many objects. For one small cell, per-clique computation is affordable. However, across a plane the objects multiply twice: the number of  $r$ -cliques explodes with  $r$ , and every cell owns its own copy of the work. Even the ten-vertex graph of Figure 1 carries 61 cliques of size three to five; on web-it-2004 the count exceeds  $10^{12}$ . Therefore, any approach that touches each  $r$ -clique of each cell even once is ruled out from the start. Counting without enumeration.

Peeling is driven by counts: each removal must decrement the count of every  $r$ -clique that shared a supporting  $s$ -clique with the removed one. However,  $s$ -cliques outnumber  $r$ -cliques, and at high  $s$  they cannot even be listed once. Therefore, both the initial counts and every later decrement must be obtained without enumerating a single  $s$ -clique. A plane of cells. A bounded plane contains dozens of cells, and each cell is a full decomposition in its own right. However, running even the best single-cell algorithm once per cell multiplies its cost by the number of cells. Worse, the deep cells are exactly the ones no existing algorithm reaches. Therefore, the computation done in one cell must be made to pay for all the others.

**Our Idea.** Our solution is built on one structural observation: vertices that belong to exactly the same maximal cliques are interchangeable, and the whole plane respects this symmetry. We call the resulting grouping the *clique quotient* of the graph. Peel patterns, not cliques.

Under the quotient, an  $r$ -clique is described by how many vertices it takes from each group, not by which vertices it takes. The 33 triangles of Figure 1 collapse into 7 such patterns; on real graphs the collapse reaches six orders of magnitude. The peel runs unchanged on patterns, with every step carrying a multiplicity. Count as a coefficient. The number of  $s$ -cliques containing a pattern is a coefficient of one small polynomial derived from the quotient. Both the initial counts and all later decrements reduce to coefficient extraction, so no  $s$ -clique is ever listed. Certify instead of recompute. Two transfer theorems link neighboring cells. Along  $s$ , a pattern whose value reaches its closed form keeps that form in every later cell. Along  $r$ , the starting values of a cell follow from the cell below it. Together they leave a single cell to peel; every other cell of the plane is arithmetic.

The consequence is an index that stores almost nothing. Whenever the closed form is certified, the value is reconstructed at query time from the quotient itself, so certified patterns cost zero bytes. We call the structure NSI, for Nucleus Spectrum Index.

**Contributions.** Our key contributions are summarized as follows. The first index over the nucleus plane. We propose NSI (Section 7), the first index that answers every cell of a bounded  $(r, s)$  plane from one structure. On web-it-2004, the explicit table behind the same queries would occupy 46.2 TB, and it cannot be built at all for  $r \geq 4$ . NSI occupies 1.66 MB and answers a point query in 74

**Table 1: Notation used across sections.**

| Symbol           | Meaning                                                                 |
|------------------|-------------------------------------------------------------------------|
| $G = (V, E)$     | input graph, $n =  V $ , $m =  E $                                      |
| $r, s$           | scored-clique size and witness-clique size, $r < s$                     |
| $R, S$           | an $r$ -clique (scored) and an $s$ -clique (witness)                    |
| $\deg_{(s)}(R)$  | $s$ -clique support of $R$ (Definition 2.2)                             |
| $\kappa_s(R)$    | core number of $R$ at witness size $s$ (Definition 2.4)                 |
| $\mathcal{M}(G)$ | the maximal cliques of $G$                                              |
| $\omega(R)$      | local clique number: size of the largest clique containing $R$          |
| $\mathcal{P}$    | the indexed plane: $r \in [r_{\min}, r_{\max}]$ , $r < s \leq s_{\max}$ |

nanoseconds. A quotient framework with certified transfer. We develop the clique quotient and its peel (Sections 4–6): interchangeability, coefficient counting, and two transfer certificates that reduce the whole plane to one peeled cell. Disabling the certificates alone slows the build by 22.6 $\times$  on nasasrb, with answers unchanged. An applicability test computable in advance. We characterize exactly when the quotient compresses, by a one-number test computable before any index is built (Section 7). The test predicted every graph of our evaluation, and it also predicted the social networks the evaluation excludes. On soc-pokec the quotient compresses by a factor of only 1.001, and the test reports this without building anything. Comprehensive Experimental Evaluation. We evaluate on eight graphs from four domains, all with at least a million edges (Section 8). Against CND run serially on the same machine, our builder wins every cell of the evaluation grid, by up to 742 $\times$  in time and 226 $\times$  in memory. It also completes the deep cells on which CND exceeds every budget.

## 2 Preliminaries

In this section, we fix notation, formalize the core number of an  $r$ -clique, and state the plane-indexing problem.

**Graph notation.** We consider a simple, undirected, unweighted graph  $G = (V, E)$  with  $n = |V|$  vertices and  $m = |E|$  edges. For  $V' \subseteq V$ , we write  $G[V']$  for the subgraph of  $G$  induced by  $V'$ . Table 1 summarizes the notation used across sections; symbols local to one lemma or algorithm are introduced where they are used.

**Definition 2.1 ( $\ell$ -Clique).** For an integer  $\ell > 0$ , a vertex set  $X \subseteq V$  with  $|X| = \ell$  is an  $\ell$ -clique of  $G$  if every two distinct vertices of  $X$  are adjacent.

Following the convention of nucleus decomposition [18, 22], we call an  $r$ -clique a *scored clique*: the decomposition computes a core number for it. An  $s$ -clique is a *witness clique*: it certifies the cohesion of the scored cliques it contains.

**Maximal cliques.** A clique is *maximal* if no outside vertex can be added to it while preserving a clique;  $\mathcal{M}(G)$  denotes the set of maximal cliques of  $G$ . Every clique of  $G$  is contained in at least one maximal clique. We write  $\omega(R)$  for the size of the largest clique of  $G$  that contains the clique  $R$ , and call it the *local clique number* of  $R$ . In Figure 1,  $\omega(\{v_1, v_2, v_3\}) = 6$  by  $M_1$ ,  $\omega(\{v_7, v_8, v_9\}) = 5$  by  $M_2$ , and  $\omega(\{v_7, v_8, v_{10}\}) = 4$  by  $M_3$ .

**Definition 2.2 ( $s$ -Clique Support).** The  *$s$ -clique support* of an  $r$ -clique  $R$  in  $G$  is  $\deg_{(s)}(R) = |\{S \subseteq V : R \subseteq S, S \text{ is an } s\text{-clique of } G\}|$ .

In Figure 1,  $\deg_{(4)}(\{v_1, v_2, v_3\}) = 3$ : the three 4-cliques obtained by adding  $v_4, v_5$ , or  $v_6$ .

**Nucleus and core number.** Cohesion is certified by a family of witness cliques that sustains all of its own scored cliques [18]. For a family  $\mathcal{S}$  of  $s$ -cliques, write  $R(\mathcal{S})$  for the set of  $r$ -cliques contained in at least one member of  $\mathcal{S}$ . Write  $G[\mathcal{S}]$  for the subgraph induced by the union of the members.

**Definition 2.3 ( $k$ -( $r, s$ )-Nucleus).** A  $k$ -( $r, s$ )-nucleus of  $G$  is a subgraph  $G[\mathcal{S}]$  induced by a family  $\mathcal{S}$  of  $s$ -cliques such that (i) every  $R \in R(\mathcal{S})$  is contained in at least  $k$  members of  $\mathcal{S}$ ; (ii) any two members are linked inside  $\mathcal{S}$  by a chain of members in which consecutive cliques share at least  $r$  vertices; and (iii)  $\mathcal{S}$  is maximal under inclusion among families satisfying (i) and (ii).

Support in condition (i) is counted within the family, not in  $G$ : a nucleus must sustain its own scored cliques. Condition (ii) links consecutive witnesses through shared scored cliques, since two  $s$ -cliques share  $r$  vertices exactly when they contain a common  $r$ -clique.

**Definition 2.4 (Core Number).** The *core number*  $\kappa_s(R)$  of an  $r$ -clique  $R$  is the largest  $k$  such that  $R \in R(\mathcal{S})$  for the defining family  $\mathcal{S}$  of some  $k$ -( $r, s$ )-nucleus, and 0 if  $R$  lies in no  $s$ -clique.

The size of the scored clique is implicit in its argument, so the single notation  $\kappa_s(R)$  serves every cell of the plane. The nucleus model subsumes the classical models:  $(r, s) = (1, 2)$  yields the  $k$ -core [20] and  $(r, s) = (2, 3)$  yields the  $k$ -truss [18, 23]. In Figure 1,  $\kappa_4(\{v_1, v_2, v_3\}) = 3$ , as computed in Example 1.1.

**Problem statement (plane indexing).** A *plane*  $\mathcal{P}$  is the set of cells  $(r, s)$  with  $r_{\min} \leq r \leq r_{\max}$  and  $r < s \leq s_{\max}$ . Given  $G$  and  $\mathcal{P}$ , the problem is to build an index that, for any  $s$  and any  $r$ -clique  $R$  with  $(|R|, s) \in \mathcal{P}$ , returns  $\kappa_s(R)$ . A scored clique contained in no witness has core number 0. The algorithmic sections therefore focus on scored cliques that lie in at least one maximal clique of size at least  $s$ .

The next section recalls how the state of the art computes one cell, and measures where its cost concentrates.

### 3 The State-of-the-Art Approach

In this section, we recall the peeling skeleton that all existing algorithms instantiate, and we measure where its cost concentrates.

**The peeling skeleton.** Every existing algorithm computes one cell  $(r, s)$  by the same skeleton, shown in Algorithm 1 [18, 22]. It materializes the scored cliques, initializes their supports, and repeatedly removes a clique of minimum support. The level  $k$  never decreases, and each clique receives the level at which it is removed.

Algorithm 1 enumerates the scored cliques and allocates one mutable record per clique (Line 1). It initializes each support (Line 3) and then peels: extract a minimum (Line 6), assign the current level (Line 7), and credit every scored clique that loses a witness (Lines 8–10). The state of the art, CND [22], realizes Lines 3 and 8–10 by aggregated counting, so no witness clique is ever listed one by one. This counting is what made single cells with millions of edges practical, and we adopt it as our baseline throughout.

**Cost anatomy.** The skeleton’s remaining cost sits in Line 1: every scored clique of the cell exists as a record, and every record carries mutable peel state. Memory is therefore at least proportional to

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#### Algorithm 1: The explicit peel of existing algorithms.

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Input: Graph  $G$ , cell  $(r, s)$ 
Output:  $\kappa_s(R)$  for every  $r$ -clique  $R$  of  $G$ 
1  $\mathcal{R} \leftarrow$  all  $r$ -cliques of  $G$ ; // one record per clique
2 foreach  $R \in \mathcal{R}$  do
3    $\text{sup}(R) \leftarrow \deg_{(s)}(R)$ 
4  $k \leftarrow 0$ ;
5 while  $\mathcal{R} \neq \emptyset$  do
6    $R^* \leftarrow$  a clique of minimum  $\text{sup}$  in  $\mathcal{R}$ ;
7    $k \leftarrow \max(k, \text{sup}(R^*))$ ;  $\kappa_s(R^*) \leftarrow k$ ;
8   foreach witness  $S \supseteq R^*$  not yet destroyed do
9     foreach  $R' \subset S$  with  $R' \in \mathcal{R}$ ,  $R' \neq R^*$  do
10       $\text{sup}(R') \leftarrow \text{sup}(R') - 1$ 
11   remove  $R^*$  from  $\mathcal{R}$ ;
12 return  $\kappa_s$ 

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Table 2: Scored cliques on web-it-2004 (7.2 million edges).

| $r$ | # $r$ -cliques    | one record each |
|-----|-------------------|-----------------|
| 3   | 338,786,461       | 2.7 GB          |
| 4   | 28,530,662,583    | 228 GB          |
| 5   | 2,140,698,950,272 | 17 TB           |

the number of scored cliques, and time to the stream of per-clique updates. Both grow combinatorially with  $r$ , as Table 2 shows for web-it-2004.

The second column of Table 2 counts the objects of Line 1; the third charges a conservative eight bytes per record. At  $r = 4$  the records alone exceed the memory of a large server, and at  $r = 5$  they exceed it by two orders of magnitude. The measured behavior matches: on web-it-2004, CND spends 5,581 seconds and 101 GB on the single shallow cell  $(3, 4)$ . Within a 1,800-second, 300 GB budget, it finishes none of the nine cells of our evaluation grid (Section 8). **What a plane demands.** The skeleton is also strictly per cell: nothing computed at  $(r, s)$  is reused at  $(r, s+1)$  or  $(r+1, s)$ . A plane therefore multiplies the cost by its number of cells. Aggregated counting removed the witness side of the cost; the scored side, one record per clique per cell, remains. Removing it requires a representation in which many scored cliques share one record and all cells share one representation. Building that representation is the subject of the next section.

### 4 The Clique Quotient

Algorithm 1 pays once per scored clique, so its cost is fixed by how many scored cliques exist. This section asks when two scored cliques are genuinely different, and shows that on real graphs most of them are not. We first group vertices, then group cliques, and finally quantify the collapse.

**Classes.** For a vertex  $v$ , let  $\mathcal{M}(v) = \{M \in \mathcal{M}(G) : v \in M\}$  denote the maximal cliques that contain it.

**Definition 4.1 (Class).** A *class* of  $G$  is a maximal set of vertices that all have the same maximal-clique membership:  $u$  and  $v$  belong to one class if and only if  $\mathcal{M}(u) = \mathcal{M}(v)$ .

Classes partition  $V$ , and the number of classes never exceeds  $n$ . In Figure 1, the classes are  $A = \{v_1, \dots, v_4\}$  inside  $M_1$  only;  $B = \{v_5, v_6\}$  shared by  $M_1$  and  $M_2$ ;  $C = \{v_7, v_8, v_9\}$  shared by  $M_2$  and  $M_3$ ; and  $D = \{v_{10}\}$  inside  $M_3$  only. The vertex colors of the figure encode

exactly this partition. Classes sit cleanly inside maximal cliques, which the following lemma records.

**LEMMA 4.2 (CLASS CONTAINMENT).** *Let  $c$  be a class and  $M$  a maximal clique. Then either every vertex of  $c$  belongs to  $M$ , or none does.*

**PROOF.** Suppose  $u \in c$  lies in  $M$ , and let  $v \in c$  be any other vertex of the class. Then  $M \in \mathcal{M}(u) = \mathcal{M}(v)$ , so  $v \in M$ .  $\square$

**Interchangeability.** Vertices of one class are not merely similar; they are indistinguishable by the graph itself.

**THEOREM 4.3 (INTERCHANGEABILITY).** *Let  $u \neq v$  be vertices of one class. Then the transposition that swaps  $u$  and  $v$  and fixes every other vertex is an automorphism of  $G$ . Consequently, for every  $s$  and every  $r$ -clique  $R$  with  $u \in R$  and  $v \notin R$ ,  $\kappa_s(R) = \kappa_s(R \setminus \{u\} \cup \{v\})$ .*

**PROOF.** Let  $M$  be any maximal clique containing  $u$ ; since  $\mathcal{M}(u) = \mathcal{M}(v)$ , it also contains  $v$ , so  $u$  and  $v$  are adjacent. Let  $w$  be any neighbor of  $u$  other than  $v$ . The edge  $\{u, w\}$  lies in some maximal clique  $M'$ , and  $M' \in \mathcal{M}(u) = \mathcal{M}(v)$  places  $v$  in  $M'$ , so  $w$  is also a neighbor of  $v$ . Hence  $N[u] \subseteq N[v]$ , and by symmetry  $N[u] = N[v]$ . A transposition of two adjacent vertices with equal closed neighborhoods preserves adjacency, so it is an automorphism. Core numbers depend only on the isomorphism type of  $G$ , so every automorphism preserves  $\kappa_s$ . Applying the transposition to  $R$  yields  $R \setminus \{u\} \cup \{v\}$ , which proves the claim.  $\square$

In Figure 1, swapping  $v_1$  and  $v_2$  maps the graph onto itself, so every core number that mentions  $v_1$  has a twin that mentions  $v_2$ . **Patterns.** Interchangeability licenses forgetting which vertices a scored clique uses.

**Definition 4.4 (Pattern).** The *pattern* of an  $r$ -clique  $R$  is the multiset of classes of its vertices, written  $P = \{c_1^{b_1}, \dots, c_t^{b_t}\}$  with  $b_1 + \dots + b_t = r$ , where  $R$  takes  $b_i$  vertices from class  $c_i$ .

**COROLLARY 4.5.** *Two  $r$ -cliques with the same pattern have the same core number at every  $s$ .*

**PROOF.** Permuting vertices within one class is a product of same-class transpositions, each an automorphism by Theorem 4.3, so it is an automorphism. For two cliques with one pattern, some within-class permutation maps the first onto the second. Automorphisms preserve  $\kappa_s$ , which proves the claim.  $\square$

Core numbers are therefore functions of patterns, and we write  $\kappa_s(P)$  for the shared value. The peel of Algorithm 1 can run on patterns directly, provided each pattern knows how many cliques it stands for.

**LEMMA 4.6 (PATTERN MULTIPLICITY).** *Let  $P = \{c_1^{b_1}, \dots, c_t^{b_t}\}$  be the pattern of at least one  $r$ -clique, and let  $n_i = |c_i|$ . Then the number of  $r$ -cliques with pattern  $P$  is exactly  $\prod_{i=1}^t \binom{n_i}{b_i}$ .*

**PROOF.** Some  $r$ -clique with pattern  $P$  lies in a maximal clique  $M$ , so each  $c_i$  intersects  $M$ , and Lemma 4.2 puts each  $c_i$  wholly inside  $M$ . All vertices of  $c_1 \cup \dots \cup c_t$  are then pairwise adjacent, so every choice of  $b_i$  vertices from each  $c_i$  forms an  $r$ -clique, and each such clique has pattern  $P$ . The number of choices is  $\prod_i \binom{n_i}{b_i}$ .  $\square$

**Table 3: The patterns of  $G$  at  $r = 3$ .**

| pattern    | multiplicity                | #cliques |
|------------|-----------------------------|----------|
| $\{A^3\}$  | $\binom{4}{3}$              | 4        |
| $\{A^2B\}$ | $\binom{4}{2} \binom{2}{1}$ | 12       |
| $\{AB^2\}$ | $\binom{4}{1} \binom{2}{2}$ | 4        |
| $\{B^2C\}$ | $\binom{2}{2} \binom{3}{1}$ | 3        |
| $\{BC^2\}$ | $\binom{2}{1} \binom{3}{2}$ | 6        |
| $\{C^3\}$  | $\binom{3}{3}$              | 1        |
| $\{C^2D\}$ | $\binom{3}{2} \binom{1}{1}$ | 3        |
| 7 patterns |                             | 33       |

**The collapse, quantified.** Summing Lemma 4.6 over the realized patterns counts all scored cliques. The ratio of cliques to patterns is therefore the average multiplicity:

$$\frac{\#r\text{-cliques}}{\#\text{patterns}} = \text{avg}_P \prod_i \binom{n_i}{b_i}. \quad (1)$$

The ratio is driven by class sizes: one class of hundreds of vertices contributes binomials that dwarf any number of singleton classes.

**Example 4.7.** Consider the graph  $G$  in Figure 1 at  $r = 3$ . Its 33 triangles realize the 7 patterns of Table 3; for instance, all 12 triangles with pattern  $\{A^2B\}$  share the core number computed once in Example 1.1. Across the whole plane  $r \in [3, 5]$ ,  $s \leq 8$ , the 61 scored cliques of  $G$  realize 16 patterns.

On real graphs the collapse is far larger. On pkustk13, a finite-element mesh, 94,893 vertices form 31,872 classes, and the  $2.1 \times 10^{12}$  5-cliques of web-it-2004 realize 1.3 million patterns, a collapse of six orders of magnitude. Section 7 returns to Equation (1) as a predictor: the average is computable before any decomposition, and it decides in advance whether the quotient is worth building.

Patterns are now the unit of peeling. The peel still needs their supports, and obtaining those without enumerating witnesses is the subject of the next section.

## 5 Peeling in the Quotient

Patterns are the peel unit, but the peel is driven by supports, and supports count witnesses. This section computes every support, and every update to it, without listing a single witness clique. We first partition the witnesses into boxes, then express support as one coefficient, and then assemble the cell algorithm PeelCell.

**Boxes.** Inside one maximal clique, counting is trivial. All classes of  $M_1$  in Figure 1 lie wholly inside it (Lemma 4.2), so the witnesses inside  $M_1$  are counted by binomials over class sizes. Across maximal cliques, however, witnesses of overlaps would be counted twice. The remedy is a classical pivoting argument [7, 22], lifted from vertices to classes: the witnesses split into disjoint groups, each with a product structure.

**Definition 5.1 (Box).** A *box*  $B$  is a set of classes  $C_B$ , together with per-class bounds  $\ell_c \leq u_c \leq n_c$  for  $c \in C_B$ , where  $n_c = |c|$ . A clique *belongs to*  $B$  if it uses  $y_c$  vertices of each class  $c \in C_B$  with  $\ell_c \leq y_c \leq u_c$ , and no vertex outside  $C_B$ .

**LEMMA 5.2 (BOX PARTITION).** *From  $\mathcal{M}(G)$  one can compute, once for the whole plane, a family  $\mathcal{B}$  of boxes such that every clique of*

every size  $s \leq s_{\max}$  belongs to exactly one box, and the classes of each box are pairwise adjacent.

The construction runs a pivoted recursion over the class graph and is deferred to Appendix A.1; only the two properties above are used in the sequel.

*Example 5.3.* Consider the graph  $G$  in Figure 1 with  $s_{\max} = 8$ . Three boxes suffice:  $B_1 = \{A, B\}$  with free bounds,  $B_2 = \{B, C\}$  requiring at least one  $C$  vertex, and  $B_3 = \{C, D\}$  requiring at least one  $D$  vertex. The two lower bounds make the split disjoint at every size. At  $s = 4$  the boxes hold  $\binom{6}{4} = 15$ ,  $\binom{5}{4} = 5$ , and  $\binom{4}{4} = 1$  witnesses: 21 in total, and each witness in exactly one box. At  $s = 5$  the counts are 6, 1, and 0, matching the 7 witnesses of  $G$ .

**Support as one coefficient.** Fix a box  $B$  and a pattern  $P$  whose classes all lie in  $C_B$ , with composition  $b$  on those classes. During the peel, some patterns have already been removed; on box  $B$ , let  $A_B$  collect the compositions of the removed patterns hosted by  $B$ .

*Definition 5.4 (Box Coefficient).* For a box  $B$ , a composition  $b$ , a witness size  $s$ , and a set  $A_B$  of removed compositions,

$$\text{Coef}_s(B, b, A_B) = \sum_{\substack{y: \ell \vee b \leq y \leq u, \\ \exists a \in A_B: a \leq y}} \sum_{\sum_c y_c = s} \prod_{c \in C_B} \binom{n_c - b_c}{y_c - b_c}. \quad (2)$$

With  $A_B = \emptyset$ , Equation (2) is the coefficient of  $x^s$  in the polynomial  $\prod_{c \in C_B} (\sum_j \binom{n_c - b_c}{j - b_c} x^j)$ , and is computed in  $O(|C_B| \cdot s)$  time by one pass of dynamic programming. A witness is *surviving* if it contains no removed scored clique.

**LEMMA 5.5 (RESIDUAL SUPPORT).** *At any point of a peel that removes whole patterns, and for any instance  $R$  of a pattern  $P$ , the number of surviving witnesses of size  $s$  containing  $R$  equals  $\sum_B \text{Coef}_s(B, b_P, A_B)$ , summed over the boxes hosting  $P$ .*

**PROOF.** By Lemma 5.2, the witnesses containing  $R$  split disjointly over the boxes whose classes cover  $P$ , so it suffices to count within one box  $B$ . The witnesses of  $B$  containing  $R$  are the choices of  $y_c - b_c$  further vertices per class outside the  $b_c$  vertices of  $R$ , giving the product of Equation (2) for each admissible  $y$ . It remains to show a witness  $W$  with composition  $y$  is destroyed exactly when  $a \leq y$  for some removed composition  $a \in A_B$ . If  $a \leq y$ , then choosing  $a_c$  vertices of  $W$  per class yields a removed scored clique inside  $W$ , since the classes of  $B$  are pairwise adjacent. Conversely, a removed clique inside  $W$  uses at most  $y_c$  vertices per class, so its composition  $a$  satisfies  $a \leq y$ . Filtering these  $y$  out of the sum is precisely the third constraint of Equation (2).  $\square$

One formula therefore serves the whole peel: initialization is the case  $A_B = \emptyset$ , and every deletion is one insertion into  $A_B$ . The implementation never expands the filtered sum; it credits the compositions that die at each removal, as detailed in Appendix A.3.

**The cell algorithm.** Algorithm 2 assembles the pieces.

PeelCell initializes every support by the closed form (Line 2), then peels patterns exactly as Algorithm 1 peels cliques: extract a minimum (Line 5), assign the level (Line 6), record the removal in each hosting box (Line 8), and refresh the neighbors' supports through Lemma 5.5 (Line 10). No witness clique appears anywhere: Lines 2 and 10 manipulate coefficients, and Line 8 manipulates compositions.

---

**Algorithm 2:** PeelCell: peel one cell in the quotient.

---

**Input:** Boxes  $\mathcal{B}$ , patterns of size  $r$  with multiplicities, cell  $(r, s)$   
**Output:**  $\kappa_s(P)$  for every pattern  $P$

```

1 foreach pattern  $P$  do
2    $\text{sup}(P) \leftarrow \sum_B \text{Coef}_s(B, b_P, \emptyset)$ 
3  $k \leftarrow 0$ ;  $A_B \leftarrow \emptyset$  for every box  $B$ ;
4 while some pattern is unpeeled do
5    $P^* \leftarrow$  an unpeeled pattern of minimum  $\text{sup}$ ;
6    $k \leftarrow \max(k, \text{sup}(P^*))$ ;  $\kappa_s(P^*) \leftarrow k$ ;
7   foreach box  $B$  hosting  $P^*$  do
8      $\text{insert } b_{P^*}$  into  $A_B$ 
9   foreach unpeeled pattern  $Q$  sharing a box with  $P^*$  do
10     $\text{sup}(Q) \leftarrow \sum_B \text{Coef}_s(B, b_Q, A_B)$ 
11    mark  $P^*$  peeled;
12 return  $\kappa_s$ 
```

---

**THEOREM 5.6 (ORBIT PEEL).** *Algorithm 2 outputs  $\kappa_s(P) = \kappa_s(R)$  for every pattern  $P$  and every instance  $R$  of  $P$ .*

**PROOF.** The removed set is always a union of whole patterns, hence invariant under every within-class automorphism of Corollary 4.5. Such automorphisms map surviving witnesses to surviving witnesses, so at every step all instances of one pattern have equal surviving support, and Lemma 5.5 computes it. The output of the explicit peel does not depend on which minimum-support clique is removed first [18], so we may run Algorithm 1 exhausting one minimum pattern at a time. While an orbit is being exhausted, internal removals only lower the supports of its remaining instances, which stay minimal and receive the same level  $k$  by Line 7 of Algorithm 1. Between removals of other patterns, supports evolve exactly as  $A_B$  records. Hence Algorithm 2 simulates a valid run of Algorithm 1 and assigns each pattern the common core number of its instances.  $\square$

*Example 5.7.* Consider the cell  $(3, 4)$  of the graph  $G$  in Figure 1. Line 2 gives  $\text{sup}(\{A^2B\}) = 3$  on  $B_1$ ,  $\text{sup}(\{C^3\}) = 2 + 1 = 3$  across  $B_2$  and  $B_3$ , and  $\text{sup}(\{C^2D\}) = 1$ . The first extraction removes  $\{C^2D\}$  at level 1, inserting  $(2, 1)$  into  $A_{B_3}$ . The refresh of  $\{C^3\}$  now filters every  $y \geq (3, 0)$  with  $y \geq (2, 1)$  out of  $B_3$ , so its support drops to 2: the removal cascaded through a shared witness, and Equation (2) accounted for it without listing that witness. Two further rounds remove the patterns at levels 2 and 3, reproducing the values of Example 1.1 for all 33 triangles at once.

**Complexity.** Line 2 costs  $O(|C_B| \cdot s)$  per hosted pattern and box. Each removal inserts one composition per hosting box and refreshes only the patterns sharing a box, by crediting the compositions that die (Appendix A.3). The peel therefore scales with patterns and their box incidences, never with cliques. On pkustk13 at  $(3, 4)$ , PeelCell peels 2.2 million patterns in place of 45.2 million scored cliques. At  $r = 5$  the gap widens to 26.8 million against 2.28 billion.

One cell is now affordable. The next section makes one cell pay for the entire plane.

## 6 From One Cell to the Plane

PeelCell prices one cell, but the plane contains dozens, and each is a full decomposition. This section proves two transfer theorems: values travel along  $s$  within a row, and across the boundary from

one row to the next. Together they leave a single cell to peel; every other cell of the plane follows by arithmetic.

**The clique floor.** The proofs work with a support form of the core number, which the peel view of Section 3 suggests [18].

LEMMA 6.1 (MAX-MIN FORM). *For every  $r$ -clique  $R$ , the core number  $\kappa_s(R)$  equals the largest  $k$  for which some family  $\mathcal{S}$  of  $s$ -cliques covers  $R$  and every  $r$ -clique covered by  $\mathcal{S}$  lies in at least  $k$  members.*

PROOF. The defining family of a nucleus witnessing  $\kappa_s(R)$  satisfies the condition, so the maximum is at least  $\kappa_s(R)$ . Conversely, let  $\mathcal{S}$  attain the maximum  $k$ , and let  $\mathcal{S}_0$  be the members reachable from a clique through  $R$  by chains of members sharing  $r$  vertices. Two members that contain a common  $r$ -clique share  $r$  vertices, so every  $r$ -clique covered by  $\mathcal{S}_0$  has all its containing members inside  $\mathcal{S}_0$ , and its support is unchanged. Hence  $\mathcal{S}_0$  is  $k$ -admissible and  $s$ -connected, and by finiteness it extends to a maximal such family. That family defines a  $k$ -( $r, s$ )-nucleus whose scored cliques include  $R$ , so  $\kappa_s(R) \geq k$ .  $\square$

LEMMA 6.2 (CLIQUE FLOOR). *For every  $r$ -clique  $R$  and every  $s$ ,  $\kappa_s(R) \geq \binom{\omega(R)-r}{s-r}$ .*

PROOF. If  $\omega(R) < s$  the right side is 0 and there is nothing to prove. Otherwise let  $K$  be a largest clique containing  $R$  and let  $\mathcal{S}$  be all  $s$ -cliques inside  $K$ . Every  $r$ -clique covered by  $\mathcal{S}$  lies in  $K$  and is contained in exactly  $\binom{|K|-r}{s-r}$  members, and some member contains  $R$ . Lemma 6.1 gives the bound.  $\square$

Both  $\omega$  and the floor are pattern-level quantities, since automorphisms preserve them; we write  $\omega(P)$  for the shared value. In Example 1.1, every computed value equals its floor:  $6 - 3, 5 - 3$ , and  $4 - 3$ . The two theorems below explain why this is the typical situation, and turn it into a certificate.

**Ascending in  $s$ .** The transfer along a row rests on a shadow argument in the style of Kruskal and Katona [9, 11]. For a real  $x \geq a$ , write  $\binom{x}{a}$  for  $x(x-1) \cdots (x-a+1)/a!$ , and let  $g_a(m) = \binom{x}{a-1}$  where  $x$  solves  $m = \binom{x}{a}$ ; this is the Lovász form of the shadow function [13].

LEMMA 6.3 (SHADOW BOUND). *For  $s \geq r + 2$  and every  $r$ -clique  $R$ ,  $\kappa_{s-1}(R) \geq g_{s-r}(\kappa_s(R))$ .*

The proof lowers a witness family by one vertex and bounds the loss by the Kruskal-Katona theorem; it appears in Appendix A.2.

THEOREM 6.4 (ABSORPTION). *Let  $R$  be an  $r$ -clique and  $s \geq r + 2$ . If  $\kappa_{s-1}(R) = \binom{\omega(R)-r}{s-1-r}$ , then  $\kappa_s(R) = \binom{\omega(R)-r}{s-r}$ .*

PROOF. Write  $w = \omega(R)$  and  $a = s - r$ . The floor gives  $\kappa_s(R) \geq \binom{w-r}{a}$ , and for  $w < s$  both sides are 0, so assume  $w \geq s$  and suppose  $\kappa_s(R) \geq \binom{w-r}{a} + 1$ . Lemma 6.3 then gives  $\kappa_{s-1}(R) \geq g_a(\binom{w-r}{a} + 1)$ . The solution  $x(m)$  of  $\binom{x}{a} = m$  increases strictly with  $m$ , and  $x \mapsto \binom{x}{a-1}$  increases strictly for  $x > a - 1$ . At the integer point  $x = w - r \geq a$  this yields  $g_a(\binom{w-r}{a} + 1) > \binom{w-r}{a-1} = \kappa_{s-1}(R)$ , a contradiction.  $\square$

COROLLARY 6.5 (ABSORBING CHAIN). *If  $\kappa_s(R)$  equals its floor, then  $\kappa_{s'}(R)$  equals its floor for every  $s' > s$ .*

PROOF. The conclusion of Theorem 6.4 is the hypothesis at  $s + 1$  with the same  $\omega(R)$ ; induction on  $s'$  finishes the proof.  $\square$

---

### Algorithm 3: Build: compute every cell of the plane.

---

**Input:** Graph  $G$ , plane  $\mathcal{P}$  with  $r \in [r_{\min}, r_{\max}]$ ,  $s \leq s_{\max}$   
**Output:**  $\kappa_s(P)$  for every pattern  $P$  and every cell of  $\mathcal{P}$

```

1 compute  $\mathcal{M}(G)$ , the classes, and the boxes  $\mathcal{B}$ ;
2 for  $r \leftarrow r_{\min}$  to  $r_{\max}$  do
3   enumerate the patterns of size  $r$ , with multiplicities and  $\omega$ ;
4   if  $r = r_{\min}$  then
5      $\kappa_{r+1} \leftarrow \text{PeelCell}(\mathcal{B}, \text{patterns}, (r, r+1))$ ;
6   else
7     foreach pattern  $P$  with  $\min_{c, b_c \geq 1} \kappa_r(P-c) - 1 = \omega(P) - r$  do
8       certify  $\kappa_{r+1}(P) \leftarrow \omega(P) - r$ 
9       peel the rest by PeelCell, seeded with the certified values;
10    for  $s \leftarrow r + 2$  to  $s_{\max}$  do
11      foreach pattern  $P$  with  $\kappa_{s-1}(P) = \binom{\omega(P)-r}{s-1-r}$  do
12        certify  $\kappa_s(P) \leftarrow \binom{\omega(P)-r}{s-r}$ 
13        peel the rest by PeelCell, seeded with the certified values;
14 return all  $\kappa$ 
```

---

One integer comparison per pattern therefore certifies a whole cell from its left neighbor, and certification, once gained, is never lost. The comparison is also what the plane's cost hinges on. Disabling it and peeling every cell in full slows the whole build by  $22.6\times$  on nasasrb, and single cells by  $67\times$  to  $5,406\times$  (Section 8).

**Stepping in  $r$ .** The second certificate crosses rows at their boundary cells  $(r, r + 1)$  and  $(r + 1, r + 2)$ .

THEOREM 6.6 (DIAGONAL CEILING). *For every  $(r+1)$ -clique  $R^+$ ,  $\kappa_{r+2}(R^+) \leq \min_{v \in R^+} \kappa_{r+1}(R^+ \setminus \{v\}) - 1$ .*

PROOF. Let  $\mathcal{S}$  realize  $k = \kappa_{r+2}(R^+)$  in Lemma 6.1, and let  $\mathcal{S}'$  be all  $(r+1)$ -cliques contained in members of  $\mathcal{S}$ . Let  $R'$  be any  $r$ -clique covered by  $\mathcal{S}'$ , and pick a member  $W = R' \cup \{u\}$  of  $\mathcal{S}'$  containing it.  $W$  is covered by  $\mathcal{S}$ , so distinct members  $S_1, \dots, S_k$  of  $\mathcal{S}$  contain  $W$ ; write  $S_i = W \cup \{w_i\}$  with the  $w_i$  pairwise distinct and outside  $W$ . Each  $R' \cup \{w_i\}$  is an  $(r+1)$ -subset of the clique  $S_i$ , hence a member of  $\mathcal{S}'$ , and these members differ from  $W$  and from one another. Therefore  $R'$  lies in at least  $k + 1$  members of  $\mathcal{S}'$ . Since  $R^+$  lies in a member of  $\mathcal{S}$ , the family  $\mathcal{S}'$  covers every face  $R^+ \setminus \{v\}$ , and Lemma 6.1 gives  $\kappa_{r+1}(R^+ \setminus \{v\}) \geq k + 1$  for each.  $\square$

COROLLARY 6.7 (DIAGONAL CERTIFICATE). *If  $\min_{v \in R^+} \kappa_{r+1}(R^+ \setminus \{v\}) = \omega(R^+) - r$ , then  $\kappa_{r+2}(R^+) = \omega(R^+) - r - 1$ .*

PROOF. Theorem 6.6 bounds  $\kappa_{r+2}(R^+)$  from above by  $\omega(R^+) - r - 1$ , and Lemma 6.2 matches it from below.  $\square$

*Remark 6.8.* The case  $r = 1$  of Theorem 6.6 is the classical relation between the trussness of an edge and the coreness of its endpoints [6, 23]; the theorem extends it to every consecutive pair of rows.

At pattern level the faces of  $P$  are the patterns  $P - c$  with  $b_c \geq 1$ , so the certificate is again one comparison per pattern. It guards exactly the cells the absorption cannot reach, the first cell of each row. Disabling it slows the build by  $2.4\times$  to  $3.0\times$  while every other cell stays certified (Section 8).

**The plane algorithm.** Algorithm 3 assembles the plane sweep.

Build computes the shared front end once (Line 1): the same classes and boxes serve every cell of the plane. The first row is peeled cold at its boundary (Line 5). Every later boundary is filled by the diagonal certificate from the row below (Lines 7–8), and every later cell of a row by absorption from its left neighbor (Lines 11–12).

Patterns that neither certificate reaches are the *residue*; PeelCell peels them with the certified patterns entering the removal order at their known levels (Lines 9 and 13). Seeding is sound by the certification principle: where a lower and an upper bound meet, the value is known before any peeling.

*Example 6.9.* Consider the graph  $G$  of Figure 1 with the plane  $r \in [3, 5]$ ,  $s \leq 8$ : twelve cells. Build peels exactly one of them, the cell  $(3, 4)$  of Example 5.7. Every value of row 3 equals its floor, so absorption certifies the cells  $(3, 5)$  through  $(3, 8)$  outright. At the boundary  $(4, 5)$ , the pattern  $\{A^4\}$  has faces  $\{A^3\}$  with  $\kappa_4 = 3$ , so the ceiling is 2 and meets the floor  $6 - 4$ ; the same happens for every pattern of rows 4 and 5. Twelve cells, one peel.

On real graphs the residue is just as thin. On pkustk13, the boundary of row 4 leaves 1 uncertified pattern out of 8.66 million, and the boundary of row 5 leaves none out of 26.8 million. The whole plane of a graph with 3.3 million edges is one boundary peel plus arithmetic.

Everything the plane needs is now computed. The next section shows that almost none of it needs to be stored.

## 7 The Index and Its Query

## 8 Experiments

## 9 Related Work

## 10 Conclusion

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## A Deferred Constructions

### A.1 Building the Boxes

The construction lifts the pivoting recursion of [7] from vertices to classes and records hold and pivot bounds per class; details to be included in the full version of this appendix.

### A.2 Proof of the Shadow Bound

We restate Lemma 6.3: for  $s \geq r + 2$  and every  $r$ -clique  $R$ ,  $\kappa_{s-1}(R) \geq g_{s-r}(\kappa_s(R))$ .

PROOF. Let  $k = \kappa_s(R)$ , realized by the family  $\mathcal{S}$  of Lemma 6.1, and let  $\mathcal{S}'$  be all  $(s-1)$ -cliques contained in members of  $\mathcal{S}$ . Let  $R'$  be any  $r$ -clique covered by  $\mathcal{S}'$ , and consider the family  $\mathcal{E} = \{\mathcal{S} \setminus R' : \mathcal{S} \in \mathcal{S}, R' \subseteq \mathcal{S}\}$  of  $(s-r)$ -sets, with  $|\mathcal{E}| \geq k$ . For every  $(s-1-r)$ -set  $T$  in the lower shadow of  $\mathcal{E}$ , the set  $R' \cup T$  is an  $(s-1)$ -clique inside a member of  $\mathcal{S}$ , hence a member of  $\mathcal{S}'$  containing  $R'$ . The Kruskal–Katona theorem in Lovász form [9, 11, 13] bounds the shadow by  $g_{s-r}(|\mathcal{E}|) \geq g_{s-r}(k)$ , so  $R'$  lies in at least  $g_{s-r}(k)$  members of  $\mathcal{S}'$ . Since  $s - r \geq 2$ , dropping one vertex outside  $R$  from a member containing  $R$  keeps a clique containing  $R$ , so  $\mathcal{S}'$  covers  $R$ . Lemma 6.1 then gives  $\kappa_{s-1}(R) \geq g_{s-r}(k)$ .  $\square$

### A.3 Incremental Evaluation of the Filtered Coefficient

Each insertion into  $A_B$  enumerates only the compositions it newly dominates and credits the co-hosted patterns by closed-form products; details to be included in the full version of this appendix.